Lesson 26 material: The Single-Subject Versus Group Debate

10.4: The Single-Subject Versus Group "Debate"

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Changes: formatted for print; exercise answers are in the study guide.

Learning Objectives

1. Explain some of the points of disagreement between advocates of single-subject research and advocates of group research.
2. Identify several situations in which single-subject research would be appropriate and several others in which group research would be appropriate.

Single-subject research is similar to group research—especially experimental group research—in many ways. They are both quantitative approaches that try to establish causal relationships by manipulating an independent variable, measuring a dependent variable, and controlling extraneous variables. But there are important differences between these approaches too, and these differences sometimes lead to disagreements. It is worth addressing the most common points of disagreement between single-subject researchers and group researchers and how these disagreements can be resolved. As we will see, single-subject research and group research are probably best conceptualized as complementary approaches.

Data Analysis

One set of disagreements revolves around the issue of data analysis. Some advocates of group research worry that visual inspection is inadequate for deciding whether and to what extent a treatment has affected a dependent variable. One specific concern is that visual inspection is not sensitive enough to detect weak effects. A second is that visual inspection can be unreliable, with different researchers reaching different conclusions about the same set of data (Danov & Symons, 2008)[1]. A third is that the results of visual inspection—an overall judgment of whether or not a treatment was effective—cannot be clearly and efficiently summarized or compared across studies (unlike the measures of relationship strength typically used in group research).

In general, single-subject researchers share these concerns. However, they also argue that their use of the steady state strategy, combined with their focus on strong and consistent effects, minimizes most of them. If the effect of a treatment is difficult to detect by visual inspection because the effect is weak or the data are noisy, then single-subject researchers look for ways to increase the strength of the effect or reduce the noise in the data by controlling extraneous variables (e.g., by administering the treatment more consistently). If the effect is still difficult to detect, then they are likely to consider it neither strong enough nor consistent enough to be of further interest. Many single-subject researchers also point out that statistical analysis is becoming increasingly common and that many of them are using this as a supplement to visual inspection—especially for the purpose of comparing results across studies (Scruggs & Mastropieri, 2001)[2].

Turning the tables, some advocates of single-subject research worry about the way that group researchers analyze their data. Specifically, they point out that focusing on group means can be highly misleading. Again, imagine that a treatment has a strong positive effect on half the people exposed to it and an equally strong negative effect on the other half. In a traditional between-subjects experiment, the positive effect on half the participants in the treatment condition would be statistically cancelled out by the negative effect on the other half. The mean for the treatment group would then be the same as the mean for the control group, making it seem as though the treatment had no effect when in fact it had a strong effect on every single participant!

But again, group researchers share this concern. Although they do focus on group statistics, they also emphasize the importance of examining distributions of individual scores. For example, if some participants were positively affected by a treatment and others negatively affected by it, this would produce a bimodal distribution of scores and could be detected by looking at a histogram of the data. The use of within-subjects designs is another strategy that allows group researchers to observe effects at the individual level and even to specify what percentage of individuals exhibit strong, medium, weak, and even negative effects. Finally, factorial designs can be used to examine whether the effects of an independent variable on a dependent variable differ in different groups of participants (introverts vs. extraverts).

External Validity

The second issue about which single-subject and group researchers sometimes disagree has to do with **external validity**—the ability to generalize the results of a study beyond the people and specific situation actually studied. In particular, advocates of group research point out the difficulty in knowing whether results for just a few participants are likely to generalize to others in the population. Imagine, for example, that in a single-subject study, a treatment has been shown to reduce self-injury for each of two children with intellectual disabilities. Even if the effect is strong for these two children, how can one know whether this treatment is likely to work for other children with intellectual delays?

Again, single-subject researchers share this concern. In response, they note that the strong and consistent effects they are typically interested in—even when observed in small samples—are likely to generalize to others in the population. Single-subject researchers also note that they place a strong emphasis on replicating their research results. When they observe an effect with a small sample of participants, they typically try to replicate it with another small sample—perhaps with a slightly different type of participant or under slightly different conditions. Each time they observe similar results, they rightfully become more confident in the generality of those results. Single-subject researchers can also point to the fact that the principles of classical and operant conditioning—most of which were discovered using the single-subject approach—have been successfully generalized across an incredibly wide range of species and situations.

And, once again turning the tables, single-subject researchers have concerns of their own about the external validity of group research. One extremely important point they make is that studying large groups of participants does not entirely solve the problem of generalizing to other *individuals*. Imagine, for example, a treatment that has been shown to have a small positive effect on average in a large group study. It is likely that although many participants

exhibited a small positive effect, others exhibited a large positive effect, and still others exhibited a small negative effect. When it comes to applying this treatment to another large *group*, we can be fairly sure that it will have a small effect on average. But when it comes to applying this treatment to another *individual*, we cannot be sure whether it will have a small, a large, or even a negative effect. Another point that single-subject researchers make is that group researchers also face a similar problem when they study a single situation and then generalize their results to other situations. For example, researchers who conduct a study on the effect of cell phone use on drivers on a closed oval track probably want to apply their results to drivers in many other real-world driving situations. But notice that this requires generalizing from a single situation to a population of situations. Thus the ability to generalize is based on much more than just the sheer number of participants one has studied. It requires a careful consideration of the similarity of the participants *and* situations studied to the population of participants and situations to which one wants to generalize (Shadish, Cook, & Campbell, 2002)[3].

Single-Subject and Group Research as Complementary Methods

As with quantitative and qualitative research, it is probably best to conceptualize single-subject research and group research as complementary methods that have different strengths and weaknesses and that are appropriate for answering different kinds of research questions (Kazdin, 1982)[4]. Single-subject research is particularly good for testing the effectiveness of treatments on individuals when the focus is on strong, consistent, and biologically or socially important effects. It is also especially useful when the behavior of particular individuals is of interest. Clinicians who work with only one individual at a time may find that it is their only option for doing systematic quantitative research.

Group research, on the other hand, is ideal for testing the effectiveness of treatments at the group level. Among the advantages of this approach is that it allows researchers to detect weak effects, which can be of interest for many reasons. For example, finding a weak treatment effect might lead to refinements of the treatment that eventually produce a larger and more meaningful effect. Group research is also good for studying interactions between treatments and participant characteristics. For example, if a treatment is effective for those who are high in motivation to change and ineffective for those who are low in motivation to change, then a group design can detect this much more efficiently than a single-subject design. Group research is also necessary to answer questions that cannot be addressed using the single-subject approach, including questions about independent variables that cannot be manipulated (e.g., number of siblings, extraversion, culture).

Finally, it is important to understand that the single-subject and group approaches represent different research traditions. This factor is probably the most important one affecting which approach a researcher uses. Researchers in the experimental analysis of behavior and applied behavior analysis learn to conceptualize their research questions in ways that are amenable to the single-subject approach. Researchers in most other areas of psychology learn to conceptualize their research questions in ways that are amenable to the group approach. At the same time, there are many topics in psychology in which research from the two traditions have informed each other and been successfully integrated. One example is research suggesting that both animals and humans have an innate “number sense”—an awareness of how many objects or events of a particular type have they have experienced without actually having to count them (Dehaene, 2011)[5]. Single-subject research with rats and birds and group research with human infants have shown strikingly similar abilities in those populations to discriminate small numbers of objects and events. This number sense—which probably evolved long before humans did—may even be the foundation of humans’ advanced mathematical abilities.

The Principle of Converging Evidence

Now that you have been introduced to many of the most commonly used research methods in psychology it should be readily apparent that no design is perfect. Every research design has strengths and weakness. True experiments typically have high internal validity but may have problems with external validity, while non-experimental research (e.g., correlational research) often has good external validity but poor internal validity. Each study brings us closer to the truth but no single study can ever be considered definitive. This is one reason why, in science, we say there is no such thing as scientific proof, there is only scientific evidence.

While the media will often try to reach strong conclusions on the basis of the findings of one study, scientists focus on evaluating a body of research. Scientists evaluate theories not by waiting for the perfect experiment but by looking at the overall trends in a number of partially flawed studies. The idea of **converging evidence** tells us to examine the pattern of flaws running through the research literature because the nature of this pattern can either support or undermine the conclusions we wish to draw. Suppose the findings from a number of different studies were largely consistent in supporting a particular conclusion. If all of the studies were flawed in a similar way, for example, if all of the studies were correlational and contained the third variable problem and the directionality problem, this would undermine confidence in the conclusions drawn because the consistency of the outcome may simply have resulted from a particular flaw that all of the studies shared. On the other hand, if all of the studies were flawed in different ways and the weakness of some of the studies were the strength of others (the low external validity of a true experiment was balanced by the high external validity of a correlational study), then we could be more confident in our conclusions.

While there are fundamental tradeoffs in different research methods, the diverse set of approaches used by psychologists have complementary strengths that allow us to search for converging evidence. We can reach meaningful conclusions and come closer to understanding truth by examining a large number of different studies each with different strengths and weakness. If the result of a large number of studies all conducted using different designs converge on the same conclusion then our confidence in that conclusion can be increased dramatically. In science, we strive for progress, not perfection.

References

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5. Dehaene, S. (2011). *The number sense: How the mind creates mathematics* (2nd ed.). New York, NY: Oxford. ↵

10.5: Single-Subject Research (Summary)

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Subject_Research/10.05%3A_Single-Subject_Research_(Summary)
Changes: formatted for print; exercise answers are in the study guide.

Key Takeaways

- Single-subject research—which involves testing a small number of participants and focusing intensively on the behavior of each individual—is an important alternative to group research in psychology.
- Single-subject studies must be distinguished from qualitative research on a single person or small number of individuals. Unlike more qualitative research, single-subject research focuses on understanding objective behavior through experimental manipulation and control, collecting highly structured data, and analyzing those data quantitatively.
- Single-subject research has been around since the beginning of the field of psychology. Today it is most strongly associated with the behavioral theoretical perspective, but it can in principle be used to study behavior from any perspective.
- Single-subject research designs typically involve measuring the dependent variable repeatedly over time and changing conditions (e.g., from baseline to treatment) when the dependent variable has reached a steady state. This approach allows the researcher to see whether changes in the independent variable are causing changes in the dependent variable.
- In a reversal design, the participant is tested in a baseline condition, then tested in a treatment condition, and then returned to baseline. If the dependent variable changes with the introduction of the treatment and then changes back with the return to baseline, this provides strong evidence of a treatment effect.
- In a multiple-baseline design, baselines are established for different participants, different dependent variables, or different settings—and the treatment is introduced at a different time on each baseline. If the introduction of the treatment is followed by a change in the dependent variable on each baseline, this provides strong evidence of a treatment effect.
- Single-subject researchers typically analyze their data by graphing them and making judgments about whether the independent variable is affecting the dependent variable based on level, trend, and latency.
- Differences between single-subject research and group research sometimes lead to disagreements between single-subject and group researchers. These disagreements center on the issues of data analysis and external validity (especially generalization to other people).
- Single-subject research and group research are probably best seen as complementary methods, with different strengths and weaknesses, that are appropriate for answering different kinds of research questions.

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Exercises

- Practice: Find and read a published article in psychology that reports new single-subject research. (An archive of articles published in the Journal of Applied Behavior Analysis can be found at http://www.ncbi.nlm.nih.gov/pmc/journals/309/) Write a short summary of the study.
- Practice: Design a simple single-subject study (using either a reversal or multiple-baseline design) to answer the following questions. Be sure to specify the treatment, operationally define the dependent variable, decide when and where the observations will be made, and so on.
    - Does positive attention from a parent increase a child's tooth-brushing behavior?
    - Does self-testing while studying improve a student's performance on weekly spelling tests?
    - Does regular exercise help relieve depression?

- Practice: Create a graph that displays the hypothetical results for the study you designed in Exercise 1. Write a paragraph in which you describe what the results show. Be sure to comment on level, trend, and latency.
- Discussion: Imagine you have conducted a single-subject study showing a positive effect of a treatment on the behavior of a man with social anxiety disorder. Your research has been criticized on the grounds that it cannot be generalized to others. How could you respond to this criticism?
- Discussion: Imagine you have conducted a group study showing a positive effect of a treatment on the behavior of a group of people with social anxiety disorder, but your research has been criticized on the grounds that "average" effects cannot be generalized to individuals. How could you respond to this criticism?
- Practice: Redesign as a group study the study by Hall and his colleagues described at the beginning of this chapter, and list the strengths and weaknesses of your new study compared with the original study.
- Practice: The generation effect refers to the fact that people who generate information as they are learning it (e.g., by self-testing) recall it better later than do people who simply review information. Design a single-subject study on the generation effect applied to university students learning brain anatomy.

Links on the original page

- Web page (ncbi.nlm.nih.gov): http://www.ncbi.nlm.nih.gov/pmc/journals/309